In [35]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

In [36]:
```
BIA = pd.read_csv('Banana_Island.csv', index_col='Property')
```

In [37]:
```
BIA
```

Out[37]:

Property	Month	Occupancy_Days	Daily_Revenue	Exchange_Rate	Monthly_Cost
A	Month_1	18	32130	498	246884
B	Month_1	22	21560	453	205402
C	Month_1	15	72322	470	320068
A	Month_2	18	30828	528	253725
B	Month_2	30	31414	508	200717
...	...	...	...	...	...
B	Month_35	21	23482	494	191275
C	Month_35	13	88750	492	278038
A	Month_36	19	29367	543	275532
B	Month_36	28	25337	520	174345
C	Month_36	12	79879	438	362559

108 rows × 5 columns

In [38]:
```
Property_A = BIA[BIA.index == 'A'] # I extracted on property A data from the data set
```

In [39]:
```
Property_A
```

Out[39]:

Property	Month	Occupancy_Days	Daily_Revenue	Exchange_Rate	Monthly_Cost
A	Month_1	18	32130	498	246884
A	Month_2	18	30828	528	253725
A	Month_3	16	21106	532	217410
A	Month_4	25	27843	557	254739
A	Month_5	15	29944	467	237477
A	Month_6	25	33556	463	220302
A	Month_7	17	36094	422	221163
A	Month_8	17	30662	501	204976
A	Month_9	21	35940	497	190162
A	Month_10	24	49346	598	234258
A	Month_11	19	33848	535	238150
A	Month_12	20	30739	476	265785
A	Month_13	25	33886	531	256027
A	Month_14	16	37852	494	258796
A	Month_15	18	38783	593	222828
A	Month_16	18	32988	497	232231
A	Month_17	18	28745	551	267317
A	Month_18	17	32298	594	208707
A	Month_19	22	31840	485	268696
A	Month_20	24	29458	583	242397
A	Month_21	20	35087	632	282566

Property	Month	Occupancy_Days	Daily_Revenue	Exchange_Rate	Monthly_Cost
A	Month_22	22	33325	502	270255
A	Month_23	23	29418	551	275424
A	Month_24	22	34120	519	242416
A	Month_25	22	33160	544	285313
A	Month_26	21	34347	578	240153
A	Month_27	16	33395	518	265677
A	Month_28	19	33736	551	250830
A	Month_29	21	42799	558	253688
A	Month_30	19	36895	510	244352
A	Month_31	24	36941	449	278234
A	Month_32	15	32489	546	251463
A	Month_33	22	27129	485	228509
A	Month_34	20	38151	451	291353
A	Month_35	17	38032	578	213142
A	Month_36	19	29367	543	275532

In [40]:
```
Property_B = BIA[BIA.index == 'B'] # I extracted only property B data from the dataset
Property_B
```

Out[40]:

Property	Month	Occupancy_Days	Daily_Revenue	Exchange_Rate	Monthly_Cost
B	Month_1	22	21560	453	205402
B	Month_2	30	31414	508	200717
B	Month_3	22	24451	455	198872
B	Month_4	24	24071	581	205430
B	Month_5	24	26970	463	182058
B	Month_6	20	21241	496	209974
B	Month_7	22	28772	577	203238
B	Month_8	24	25075	470	223911
B	Month_9	21	22643	553	180830
B	Month_10	28	26806	434	220897
B	Month_11	25	25730	510	193352
B	Month_12	26	25704	556	188212
B	Month_13	27	21889	529	206543
B	Month_14	28	27604	558	210451
B	Month_15	27	26648	544	210721
B	Month_16	28	20290	531	166225
B	Month_17	21	27341	523	198981
B	Month_18	22	28807	588	181912
B	Month_19	23	26105	503	184240
B	Month_20	24	27480	534	204969
B	Month_21	26	25352	478	204339

Property	Month	Occupancy_Days	Daily_Revenue	Exchange_Rate	Monthly_Cost
B	Month_22	22	24325	558	182133
B	Month_23	24	27948	434	206528
B	Month_24	20	26541	538	206061
B	Month_25	28	30473	517	206038
B	Month_26	23	31719	530	190845
B	Month_27	27	26817	549	209130
B	Month_28	27	26680	477	186894
B	Month_29	22	29319	570	204290
B	Month_30	24	22849	598	194432
B	Month_31	26	26210	568	190956
B	Month_32	23	21718	558	185555
B	Month_33	24	30204	518	204972
B	Month_34	30	29696	561	206156
B	Month_35	21	23482	494	191275
B	Month_36	28	25337	520	174345

In [41]:
```
Property_C = BIA[BIA.index == 'C']
Property_C
```

Out[41]:

Property	Month	Occupancy_Days	Daily_Revenue	Exchange_Rate	Monthly_Cost
C	Month_1	15	72322	470	320068
C	Month_2	13	75761	550	358439
C	Month_3	17	77887	487	325622
C	Month_4	16	88123	431	350614
C	Month_5	12	80448	482	381548
C	Month_6	17	86687	472	307417
C	Month_7	10	74924	594	413690
C	Month_8	12	77295	585	378588
C	Month_9	12	95057	585	365630
C	Month_10	11	75896	500	346728
C	Month_11	15	73349	537	374374
C	Month_12	13	70582	556	315161
C	Month_13	11	77483	520	365784
C	Month_14	15	78348	474	328640
C	Month_15	13	74079	523	325855
C	Month_16	15	71098	482	327005
C	Month_17	16	74731	493	344250
C	Month_18	6	74388	472	325428
C	Month_19	17	78738	450	365506
C	Month_20	17	83495	496	369006
C	Month_21	18	72996	533	353531

Property	Month	Occupancy_Days	Daily_Revenue	Exchange_Rate	Monthly_Cost
C	Month_22	18	73057	485	382350
C	Month_23	11	79550	604	317887
C	Month_24	8	80116	525	362509
C	Month_25	19	87225	517	390416
C	Month_26	14	82269	499	375464
C	Month_27	14	84010	535	359335
C	Month_28	16	80043	542	326071
C	Month_29	18	76832	495	369640
C	Month_30	14	78580	476	350275
C	Month_31	9	79297	473	325738
C	Month_32	11	74630	620	362246
C	Month_33	15	74220	576	347374
C	Month_34	18	72088	446	339224
C	Month_35	13	88750	492	278038
C	Month_36	12	79879	438	362559

I calculated the mean and std of the daily revenue, monthly cost and occupancy days for each property so i can calculate how far from the mean each std is to get the best and worse performing properties

In [42]:
```
Property_A[['Daily_Revenue','Monthly_Cost', 'Occupancy_Days']].mean()
```

Out[42]:
```
Daily_Revenue      33507.694444
Monthly_Cost      246970.472222
Occupancy_Days        19.861111
dtype: float64
```

In [43]:
```
Property_A[['Daily_Revenue','Monthly_Cost', 'Occupancy_Days']].std()
```

Out[43]:
```
Daily_Revenue      4806.126113
Monthly_Cost      24649.315869
Occupancy_Days        2.987142
dtype: float64
```

In [44]:
```
Property_B[['Daily_Revenue','Monthly_Cost', 'Occupancy_Days']].mean()
```

Out[44]:
```
Daily_Revenue      26090.861111
Monthly_Cost      197802.333333
Occupancy_Days        24.527778
dtype: float64
```

In [45]:
```
Property_B[['Daily_Revenue','Monthly_Cost', 'Occupancy_Days']].std()
```

Out[45]:
```
Daily_Revenue      2937.579773
Monthly_Cost      12890.878170
Occupancy_Days        2.813093
dtype: float64
```

In [46]:
```
Property_C[['Daily_Revenue','Monthly_Cost', 'Occupancy_Days']].mean()
```

Out[46]:
```
Daily_Revenue      78450.916667
Monthly_Cost      349778.055556
Occupancy_Days        13.916667
dtype: float64
```

In [47]:
```
Property_C[['Daily_Revenue','Monthly_Cost', 'Occupancy_Days']].std()
```

Out[47]:
```
Daily_Revenue      5597.910382
Monthly_Cost      27324.930511
Occupancy_Days        3.111040
dtype: float64
```

In [48]:
```
Avg_monthly_revenueA = (Property_A['Daily_Revenue'].mean() * Property_A['Occupancy_Days'].mean()) - (Property_A['Monthly_Cost'
Avg_monthly_revenueA.round(1)
```

Out[48]:
```
np.float64(418529.6)
```

In [49]:
```
Avg_monthly_revenueB = (Property_B['Daily_Revenue'].mean() * Property_B['Occupancy_Days'].mean()) - (Property_B['Monthly_Cost'
Avg_monthly_revenueB.round(1)
```

Out[49]:
```
np.float64(442148.5)
```

In [50]:
```
Avg_monthly_revenueC = (Property_C['Daily_Revenue'].mean() * Property_C['Occupancy_Days'].mean()) - (Property_C['Monthly_Cost'
Avg_monthly_revenueC.round(1)
```

Out[50]:
```
np.float64(741997.2)
```

In [51]:
```
Yearly_revenueA = Avg_monthly_revenueA * 12
print(Yearly_revenueA.round(1))
```

```
5022354.8
```

In [52]:
```
Yearly_revenueB = Avg_monthly_revenueB * 12
print(Yearly_revenueB.round(1))
```

```
5305782.1
```

In [53]:
```
Yearly_revenueC = Avg_monthly_revenueC * 12
print(Yearly_revenueC.round(1))
```

```
8903966.4
```

In [54]:
```
Best_Property_A_daily_income= Property_A['Daily_Revenue'].mean() + Property_A['Daily_Revenue'].std()

Best_Property_A_Monthly_Cost= Property_A['Monthly_Cost'].mean() + Property_A['Monthly_Cost'].std()

Best_Property_A_occupancy= Property_A['Occupancy_Days'].mean() + Property_A['Occupancy_Days'].std()

print(Best_Property_A_daily_income)
print(Best_Property_A_Monthly_Cost)
print(Best_Property_A_occupancy)
```

```
38313.82055788071
271619.7880915882
22.848252867542477
```

In [55]:
```
Best_Property_B_daily_income= Property_B['Daily_Revenue'].mean() + Property_B['Daily_Revenue'].std()

Best_Property_B_Monthly_Cost= Property_B['Monthly_Cost'].mean() + Property_B['Monthly_Cost'].std()

Best_Property_B_occupancy= Property_A['Occupancy_Days'].mean() + Property_B['Occupancy_Days'].std()

print(Best_Property_B_daily_income)
print(Best_Property_B_Monthly_Cost)
print(Best_Property_B_occupancy)
```

```
29028.44088415828
210693.2115035745
22.674204082111547
```

In [56]:
```
Best_Property_C_daily_income= Property_A['Daily_Revenue'].mean() + Property_C['Daily_Revenue'].std()

Best_Property_C_Monthly_Cost= Property_A['Monthly_Cost'].mean() + Property_C['Monthly_Cost'].std()

Best_Property_C_occupancy= Property_A['Occupancy_Days'].mean() + Property_C['Occupancy_Days'].std()

print(Best_Property_C_daily_income)
print(Best_Property_C_Monthly_Cost)
print(Best_Property_C_occupancy)
```

```
39105.60482689967
274295.4027331201
22.972151359737193
```

In [57]:
```
monthly_bestA = (Best_Property_A_daily_income * Best_Property_A_occupancy) - Best_Property_A_Monthly_Cost
Yearly_bestA = monthly_bestA * 12
print(Yearly_bestA.round(1))
```

```
7245408.9
```

In [58]:
```
monthly_bestB = (Best_Property_B_daily_income * Best_Property_B_occupancy) - Best_Property_B_Monthly_Cost
Yearly_bestB = monthly_bestB * 12
print(Yearly_bestB.round(1))
```

```
5370043.0
```

In [59]:
```
monthly_bestC = (Best_Property_C_daily_income * Best_Property_C_occupancy) - Best_Property_C_Monthly_Cost
Yearly_bestC = monthly_bestC * 12
print(Yearly_bestC.round(1))
```

```
7488533.6
```

In [60]:
```
worse_Property_A_daily_income= Property_A['Daily_Revenue'].mean() - Property_A['Daily_Revenue'].std()

worse_Property_A_Monthly_Cost= Property_A['Monthly_Cost'].mean() - Property_A['Monthly_Cost'].std()

worse_Property_A_occupancy= Property_A['Occupancy_Days'].mean() - Property_A['Occupancy_Days'].std()

print(worse_Property_A_daily_income)
print(worse_Property_A_Monthly_Cost)
print(worse_Property_A_occupancy)
```

```
28701.56833100818
222321.15635285623
16.873969354679744
```

In [61]:
```
worse_Property_B_daily_income= Property_B['Daily_Revenue'].mean() - Property_B['Daily_Revenue'].std()

worse_Property_B_Monthly_Cost= Property_B['Monthly_Cost'].mean() - Property_B['Monthly_Cost'].std()

worse_Property_B_occupancy= Property_B['Occupancy_Days'].mean() - Property_B['Occupancy_Days'].std()

print(worse_Property_B_daily_income)
print(worse_Property_B_Monthly_Cost)
print(worse_Property_B_occupancy)
```

```
23153.281338063938
184911.45516309218
21.714684806777342
```

In [62]:
```
worse_Property_C_daily_income= Property_C['Daily_Revenue'].mean() - Property_C['Daily_Revenue'].std()

worse_Property_C_Monthly_Cost= Property_C['Monthly_Cost'].mean() - Property_C['Monthly_Cost'].std()

worse_Property_C_occupancy= Property_C['Occupancy_Days'].mean() - Property_C['Occupancy_Days'].std()
```

```
print(worse_Property_C_daily_income)
print(worse_Property_C_Monthly_Cost)
print(worse_Property_C_occupancy)
```

```
72853.00628421144
322453.1250446577
10.805626418040585
```

In [63]:
```
monthly_worseA = (worse_Property_A_daily_income * worse_Property_A_occupancy) - worse_Property_A_Monthly_Cost
Yearly_worseA = monthly_worseA * 12
Yearly_worseA.round(1)
```

Out[63]:
```
np.float64(3143858.7)
```

In [64]:
```
monthly_worseB = (worse_Property_B_daily_income * worse_Property_B_occupancy) - worse_Property_B_Monthly_Cost
Yearly_worseB = monthly_worseB * 12
Yearly_worseB.round(1)
```

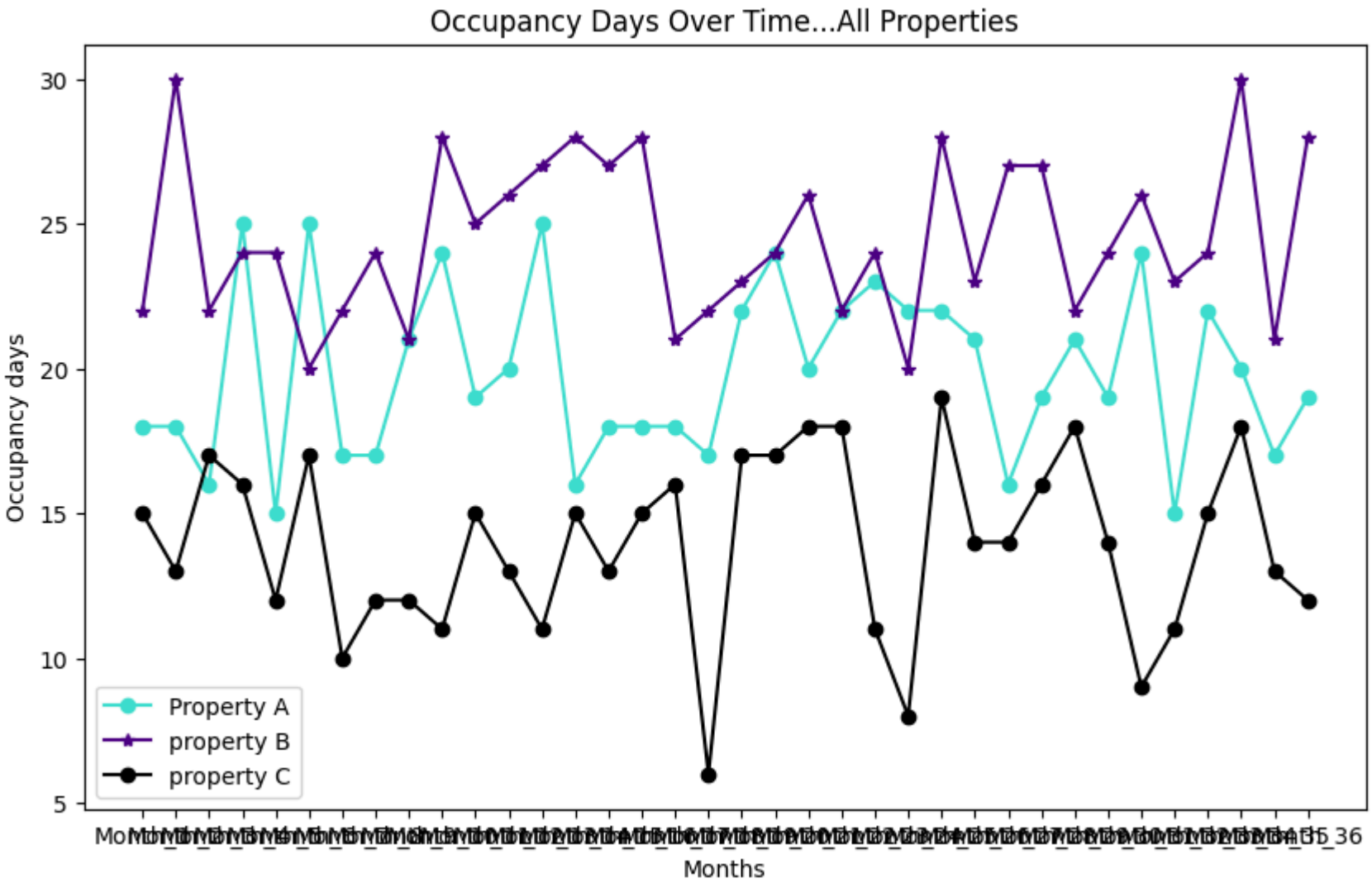
Out[64]:
```
np.float64(3814257.0)
```

In [65]:
```
monthly_worseC = (worse_Property_C_daily_income * worse_Property_C_occupancy) - worse_Property_C_Monthly_Cost
Yearly_worseC = monthly_worseC * 12
print(Yearly_worseC.round(1))
```

```
5577230.9
```

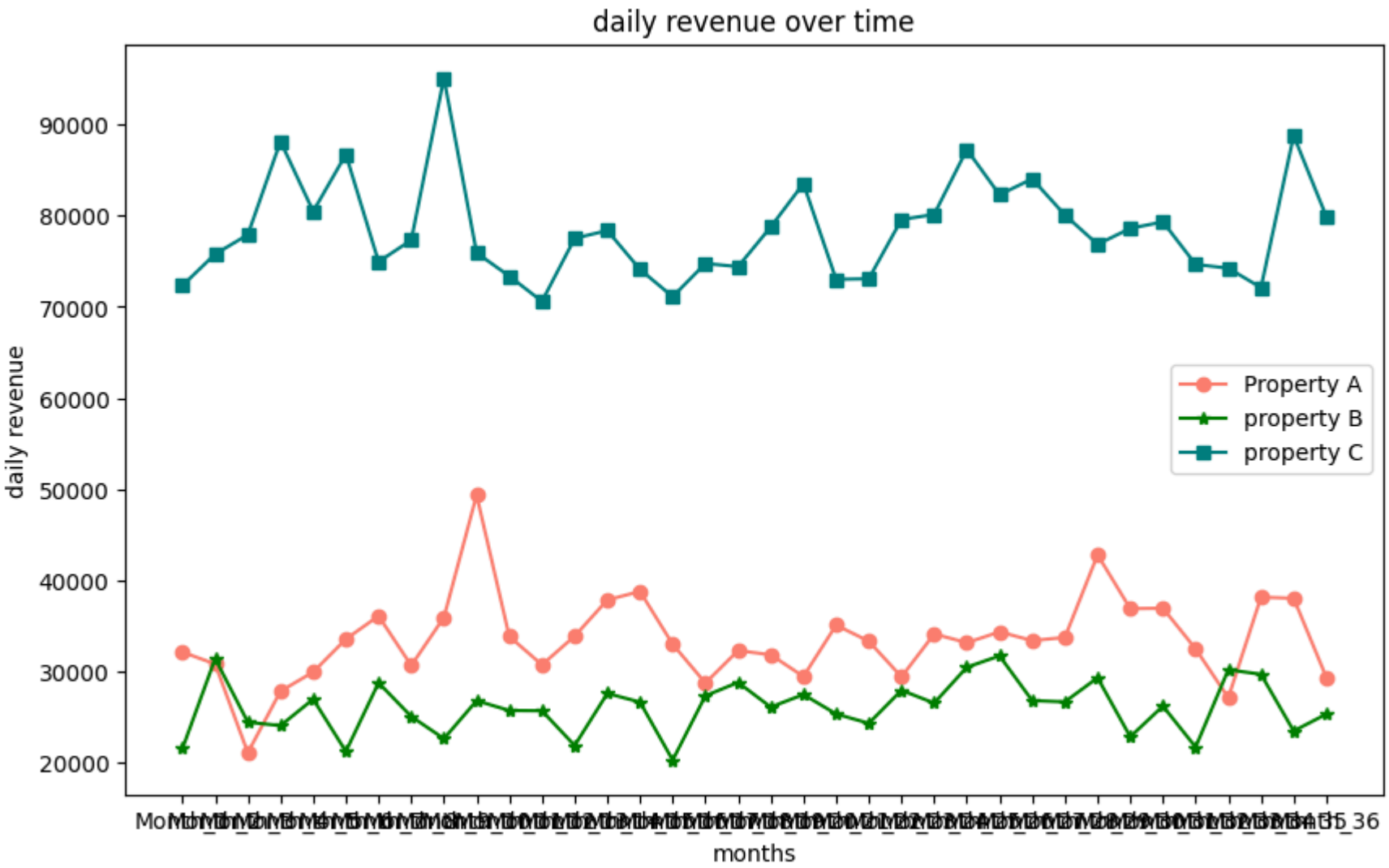
I made charts to show the trend of each property's occupancy days, daily revenue and monthly cost over time

In [67]:
```
plt.figure(figsize=(10, 6))
plt.plot(Property_A['Month'], Property_A['Occupancy_Days'], marker='o', label='Property A', color='turquoise')
plt.plot(Property_B['Month'], Property_B['Occupancy_Days'], marker='*', label='property B', color='indigo')
plt.plot(Property_C['Month'], Property_C['Occupancy_Days'], marker='o', label='property C', color= 'k')
plt.xlabel('Months')
plt.ylabel('Occupancy days')
plt.title('Occupancy Days Over Time...All Properties')
plt.legend()
plt.show()
```



In [69]:
```
plt.figure(figsize=(10, 6))
plt.plot(Property_A['Month'], Property_A['Daily_Revenue'], marker='o', label='Property A', color='salmon')
plt.plot(Property_B['Month'], Property_B['Daily_Revenue'], marker='*', label='property B', color='green')
plt.plot(Property_C['Month'], Property_C['Daily_Revenue'], marker='s', label='property C', color= 'teal')
plt.xlabel('months')
plt.ylabel('daily revenue')
plt.title('daily revenue over time')
```

```
plt.legend()
plt.show()
```

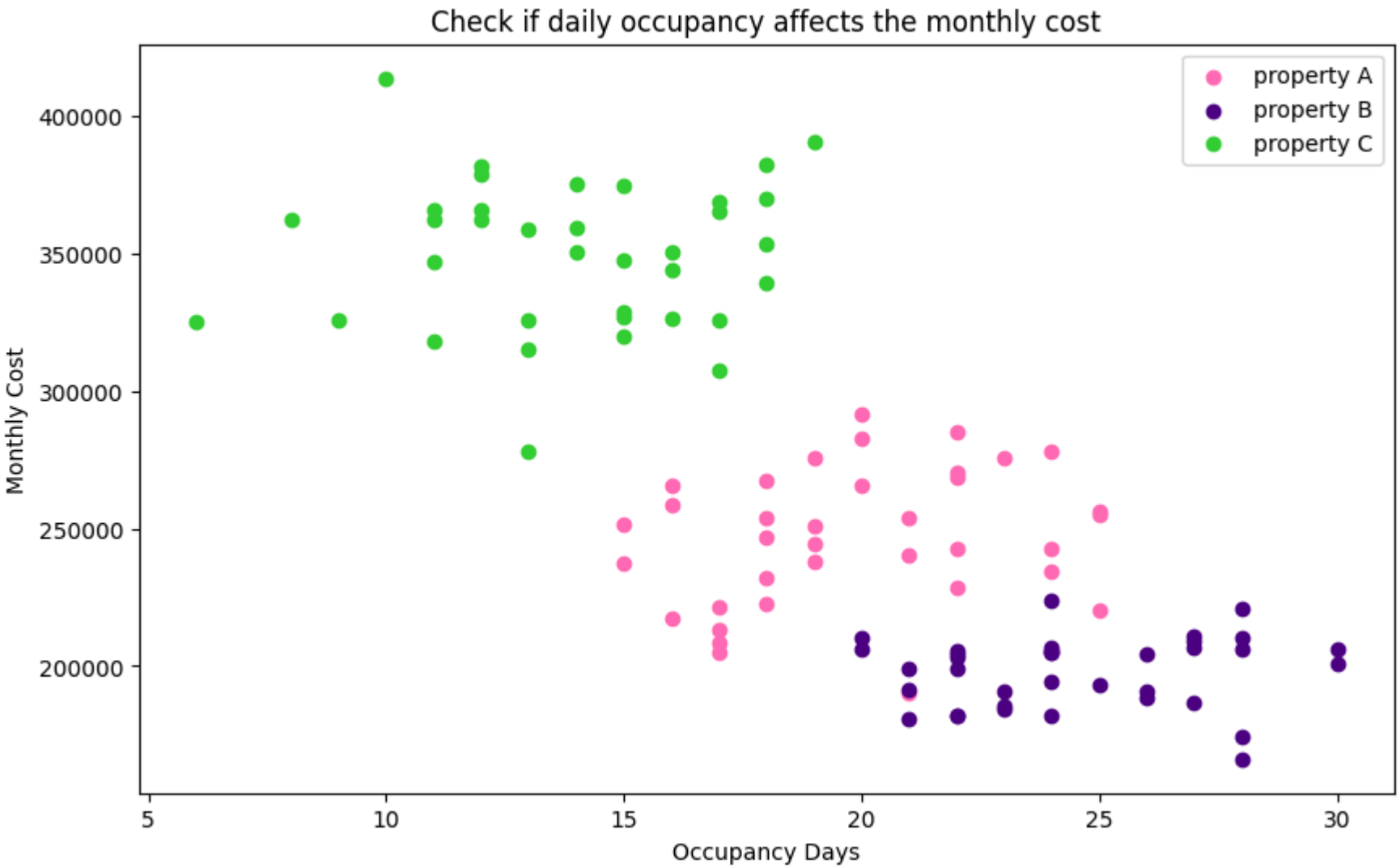


In [70]:
```
plt.figure(figsize=(10, 6))
plt.plot(Property_A['Month'], Property_A['Monthly_Cost'], marker='o', label='Property A', color='blue')
plt.plot(Property_B['Month'], Property_B['Monthly_Cost'], marker='*', label='property B', color='plum')
plt.plot(Property_C['Month'], Property_C['Monthly_Cost'], marker='s', label='property C', color= 'hotpink')
plt.xlabel('months')
```

```
plt.ylabel('montly cost')
plt.title('monthly cost over time.... over time')
plt.legend()
plt.show()
```

In [71]:
```
plt.figure(figsize=(10, 6))
plt.scatter(Property_A['Occupancy_Days'], Property_A['Monthly_Cost'], color='hotpink', label='property A')
plt.scatter(Property_B['Occupancy_Days'], Property_B['Monthly_Cost'], color='indigo', label='property B')
```

```
plt.scatter(Property_C['Occupancy_Days'], Property_C['Monthly_Cost'], color='limegreen', label='property C')
plt.xlabel('Occupancy Days')
plt.ylabel('Monthly Cost')
plt.title('Check if daily occupancy affects the monthly cost')
plt.legend()
plt.show()
```



In [72]:
```
aff_a= Property_A['Occupancy_Days'].corr(Property_A['Monthly_Cost'])
aff_b= Property_B['Occupancy_Days'].corr(Property_B['Monthly_Cost'])
aff_c= Property_C['Occupancy_Days'].corr(Property_C['Monthly_Cost'])
```

In [73]:
```
aff_a
```

Out[73]:
```
np.float64(0.23257725507930613)
```

In [75]:
```
aff_b
```

Out[75]:
```
np.float64(0.08297711814003562)
```

In [77]:
```
aff_c
```

Out[77]:
```
np.float64(0.043145760757868926)
```